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CANADIAN FEDERAL PROCUREMENT ANALYSIS

Competition, Vendor Concentration & Spending Risk

An independent analysis of $738 billion in Government of Canada contract spending, FY2015 – FY2026, built on the Open Government Portal proactive-disclosure dataset.

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THE BUSINESS PROBLEM

Where does $738B in federal spending actually go — and who is watching how it's awarded?

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No consolidated view

Contract data is published, but no single tool shows competition levels, vendor concentration, or risk patterns across departments.

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Billions in discretion

Sole-sourced and standing-offer awards involve significant department-level judgment — and carry the highest fraud/inefficiency risk.

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Real oversight stakes

Auditors, MPs, journalists and taxpayers all need this picture — but today it requires manually wrangling a 300K+ row raw file.

Research question: How competitive and concentrated is federal procurement, and where are the highest-risk spending patterns?

METHODOLOGY

A three-stage analytics pipeline

01

Clean

Python (pandas + RapidFuzz)

Standardized 200+ raw vendor-name variants, parsed fiscal years (Apr–Mar), flagged competitive vs. sole-source awards.

→

02

Aggregate

SQL (DuckDB)

Wrote analytical queries for concentration (HHI), vendor rankings, and department-level KPIs against the cleaned dataset.

→

03

Visualize

Power BI

Built a two-page interactive dashboard — an executive summary and an operational deep-dive — with a shared design system.

Scope: all federal contracts ≥ $10,000, all departments and agencies, FY2015 – FY2026 · Source: Open Government Portal — Proactive Publication of Contracts (Treasury Board of Canada Secretariat)

THE DELIVERABLE

A two-page dashboard built for two audiences

Page 1 — Executive Summary

For: leadership, oversight sponsor, quick-read audience

- Procurement strategy breakdown by department
- Market concentration index (HHI)
- Top 10 vendors by contract value
- Monthly award-volume trend

Page 2 — Operational Efficiency & Compliance

For: audit staff, CFOs, analysts wanting the detail

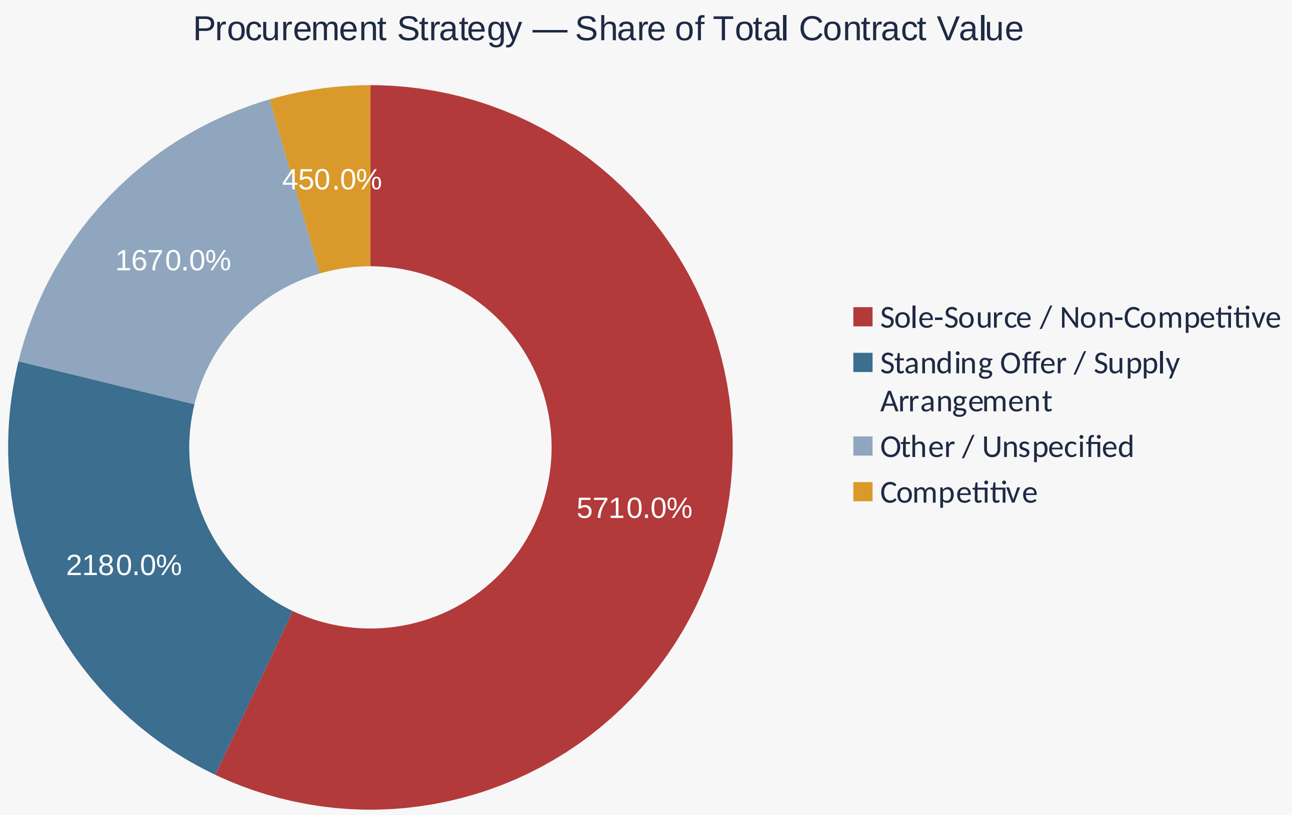
- Trade-agreement coverage (exempt vs. covered)
- Award-to-delivery efficiency by department
- Regional vendor distribution
- Progress toward the 5% Indigenous set-aside target

Just 4.5% of contract value was openly competitive

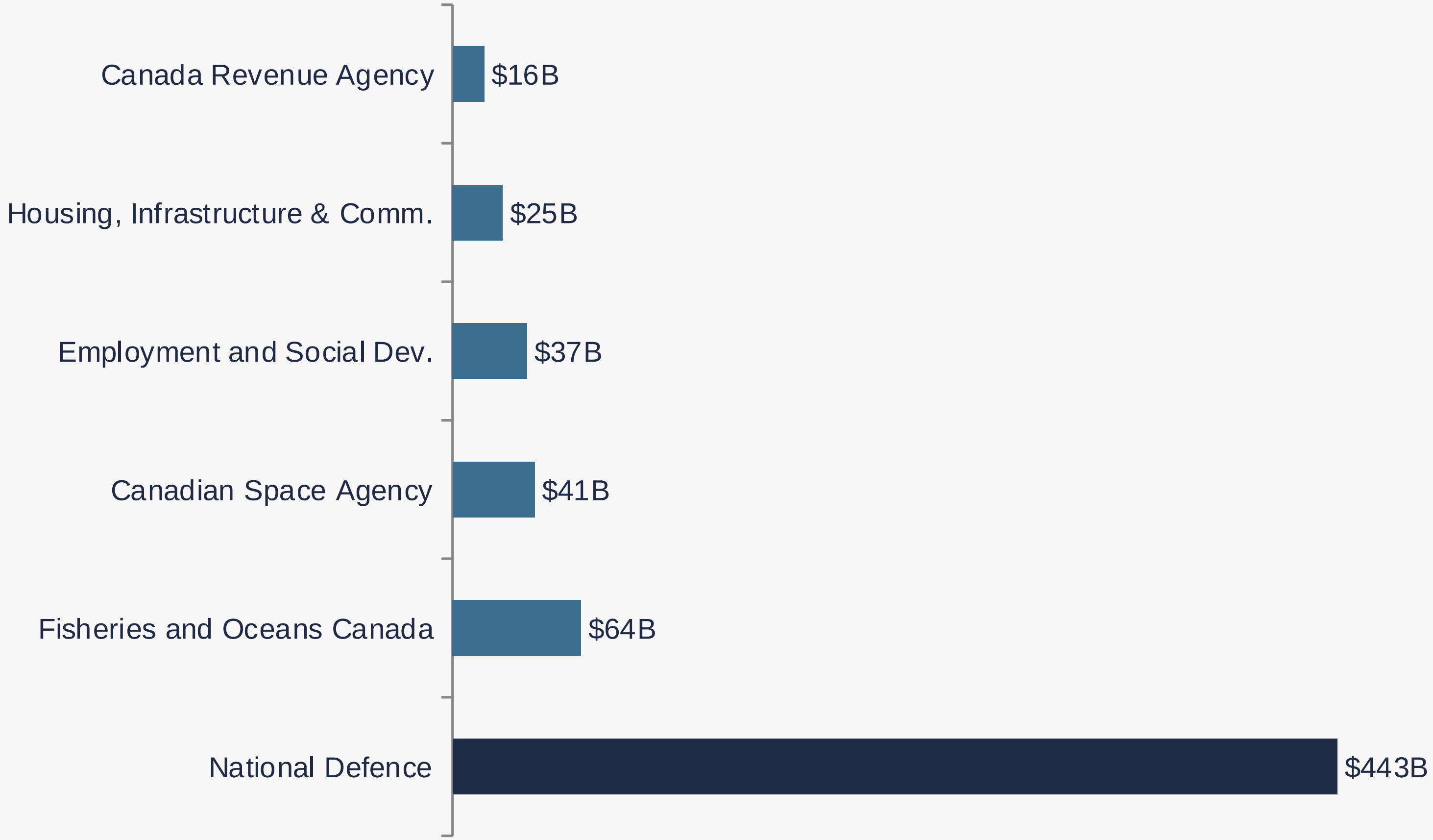
57%

of all federal contract value (FY2015–FY2026) was awarded sole-source / non-competitive

$738.3B total contract value analyzed across all departments



One department accounts for 60% of all federal contract value



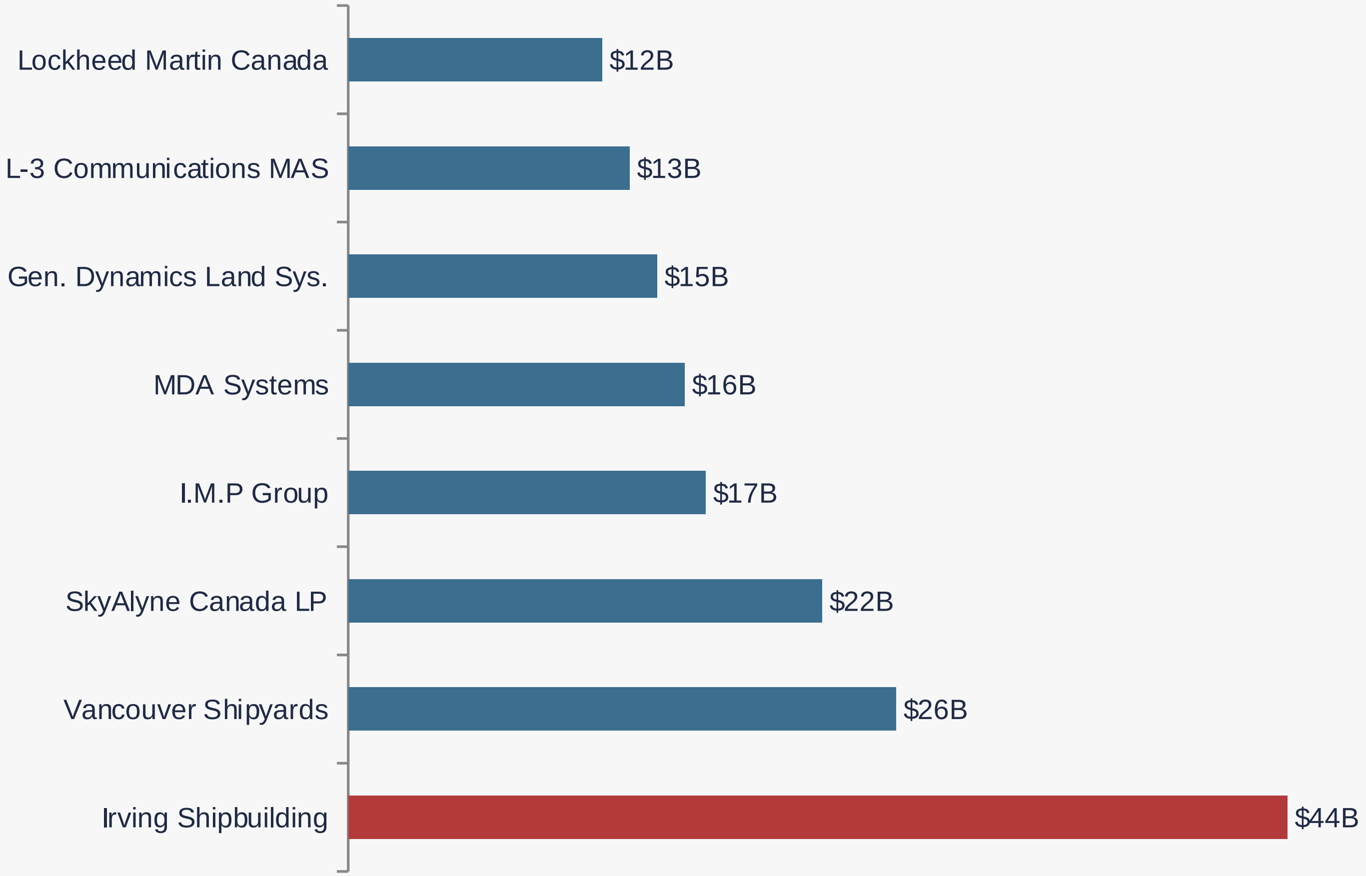
$442.8B

awarded to National Defence alone — more than the next five departments combined.

Why it matters:

Large-platform defence procurement (shipbuilding, aircraft) concentrates enormous discretionary spend in a single portfolio — making its competition practices disproportionately important to overall government risk.

Vendor concentration — and a data-cleaning catch worth flagging



METHODOLOGY NOTE

The raw data listed “Irving Shipbuilding Inc” and “Irving Shipbuilding Inc.” as two separate vendors.

Once standardized, Irving becomes the #1 vendor at $44.4B — not Vancouver Shipyards.

This is why vendor-name cleaning (Python + RapidFuzz) isn't a formality — an uncleaned dataset silently understates real vendor concentration.

Avg. department HHI: 757.6 (competitive) — but 5 departments exceed 2,500 (highly concentrated), incl. Canadian Space Agency and Housing, Infrastructure & Communities.

The Indigenous procurement target is being missed — by a wide margin

91%

of departments (65 of 71) fall below the federal government's mandatory 5% Indigenous set-aside target (PSIB).

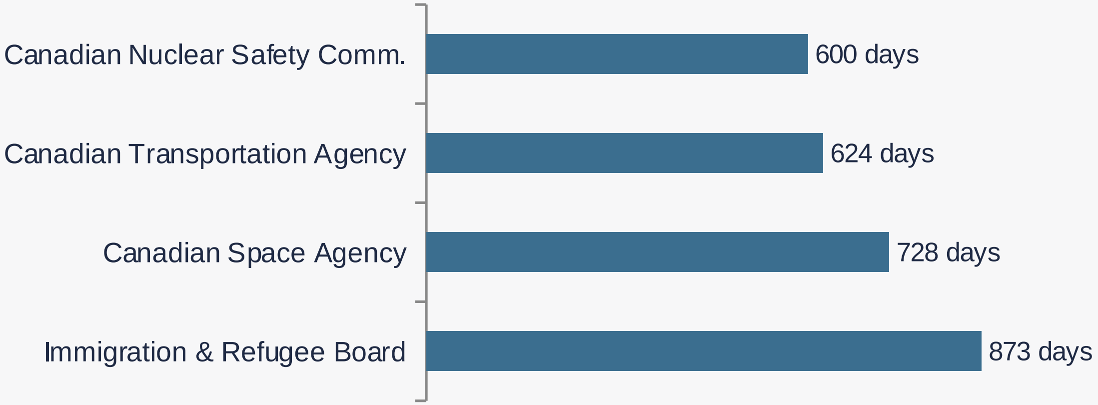
0.28%

is the government-wide, dollar-weighted average share of contract value going to Indigenous businesses — versus a 5% target.

Best performer: Elections Canada at 7.2% (exceeds target) · Many large-spend departments sit well under 2%

Delivery speed and trade-agreement coverage vary sharply by department

SLOWEST AWARD-TO-DELIVERY TIMES

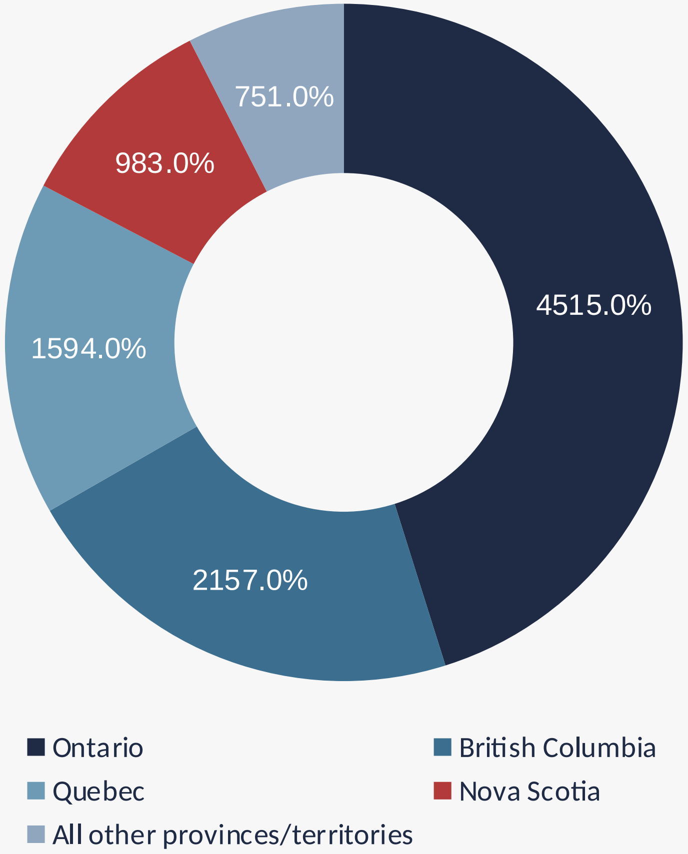


Median days are consistently lower than the average across departments — a small number of long-tail contracts (multi-year builds, R&D) are skewing the mean upward.

HIGHEST TRADE-AGREEMENT-EXEMPT SPEND

“Exempt” means the contract falls outside CFTA/CETA/etc. coverage — typically lower external competitive-bidding requirements. Ranked by exempt dollar value, not percentage, to filter out small-office noise.

Vendor location skews heavily toward shipbuilding hubs, not just Ottawa



Insight

Nova Scotia's 9.8% share is disproportionate to its population — driven almost entirely by Irving Shipbuilding's Halifax-based National Shipbuilding Strategy contracts.

Likewise, British Columbia's 21.6% is anchored by Vancouver Shipyards.

This cross-references directly with the vendor-concentration finding (Slide 07) — a handful of mega-contracts, not broad regional policy, explain the geographic pattern.

What this analysis can — and can't — claim

1 **Unaudited, self-reported data**

Treasury Board Secretariat publishes this data as-is; it has not been independently audited for accuracy or completeness.

2 **Thresholds changed over time**

The competitive-bidding threshold has shifted across the FY2015–2026 window; this analysis applies current thresholds uniformly rather than the exact rule in force on each contract date.

3 **"Sole-source" ≠ wrongdoing**

Legitimate reasons for sole-sourcing exist (national security, IP lock-in, urgency). Findings describe patterns worth review, not proven fraud.

4 **No internal justification data**

This analysis has no access to the procurement file behind each award — only the disclosed summary record.

SKILLS DEMONSTRATED

Built end-to-end with an analyst's toolkit

Python

pandas · RapidFuzz
data cleaning & entity resolution

SQL

DuckDB
aggregation & analytical queries

Power BI

DAX · Power Query
interactive 2-page dashboard

GitHub

version control
portfolio documentation

Core analytical techniques applied: fuzzy string matching for entity resolution · Herfindahl-Hirschman Index for market concentration · window-based anomaly screening · fiscal-year time series analysis

The takeaway

$738B analyzed. 57% sole-sourced. One department, 60% of spend.

Federal procurement data is public — but without analysis, its risk patterns stay invisible. This project turns a 300K+ row disclosure file into a decision-ready view of where competition, concentration, and equity targets need attention.

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